HISTOLOGY OF TONSIL AND SPLEEN

Learning Objectives

At the end of this lecture, students should be able to identify the:

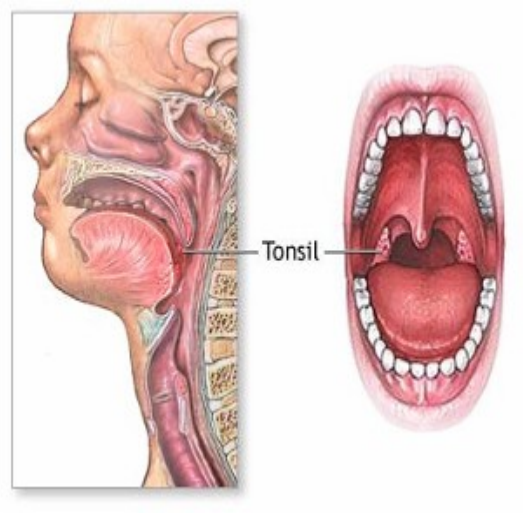
- Locations of tonsils.
- The histological features of tonsils.
- Functions of tonsils.
- Histological features of spleen.
- Functions of spleen.

Lymphoid system

- Lymphoid system gives immunity against injury by foreign substance or an organism like bacteria, viruses and parasites.

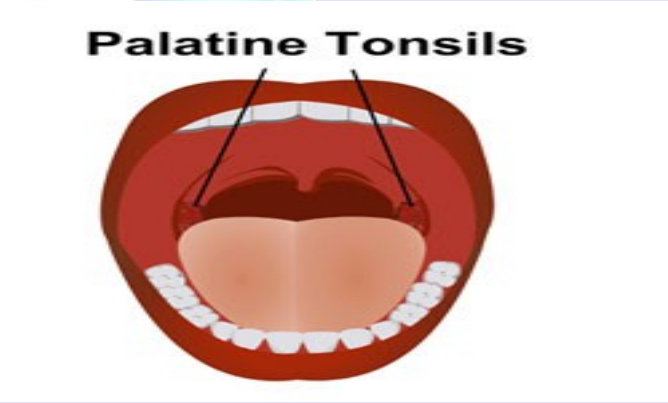
Tonsil

- These are masses of lymphoid tissue may be partially encapsulated as small organs.
- There are three such masses which are paired structures, palatine, pharyngeal and lingual tonsil.
- They all are located in the pharynx.
- **Function:**
- Production of lymphocytes.



Palatine Tonsil

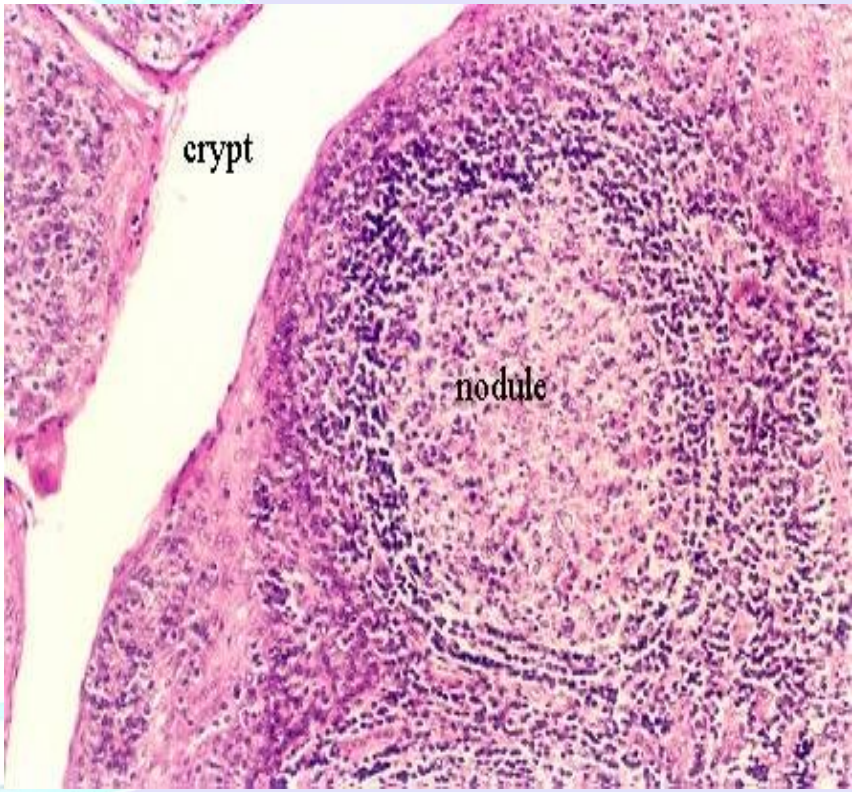
- These are paired oval shaped bodies in oropharynx between glossopharyngeal and palatopharyngeal arches (ant & post. faucial pillars).
- Free surface is covered by stratified squamous epithelium which is continuous with the epithelium of pharynx.



Palatine Tonsil

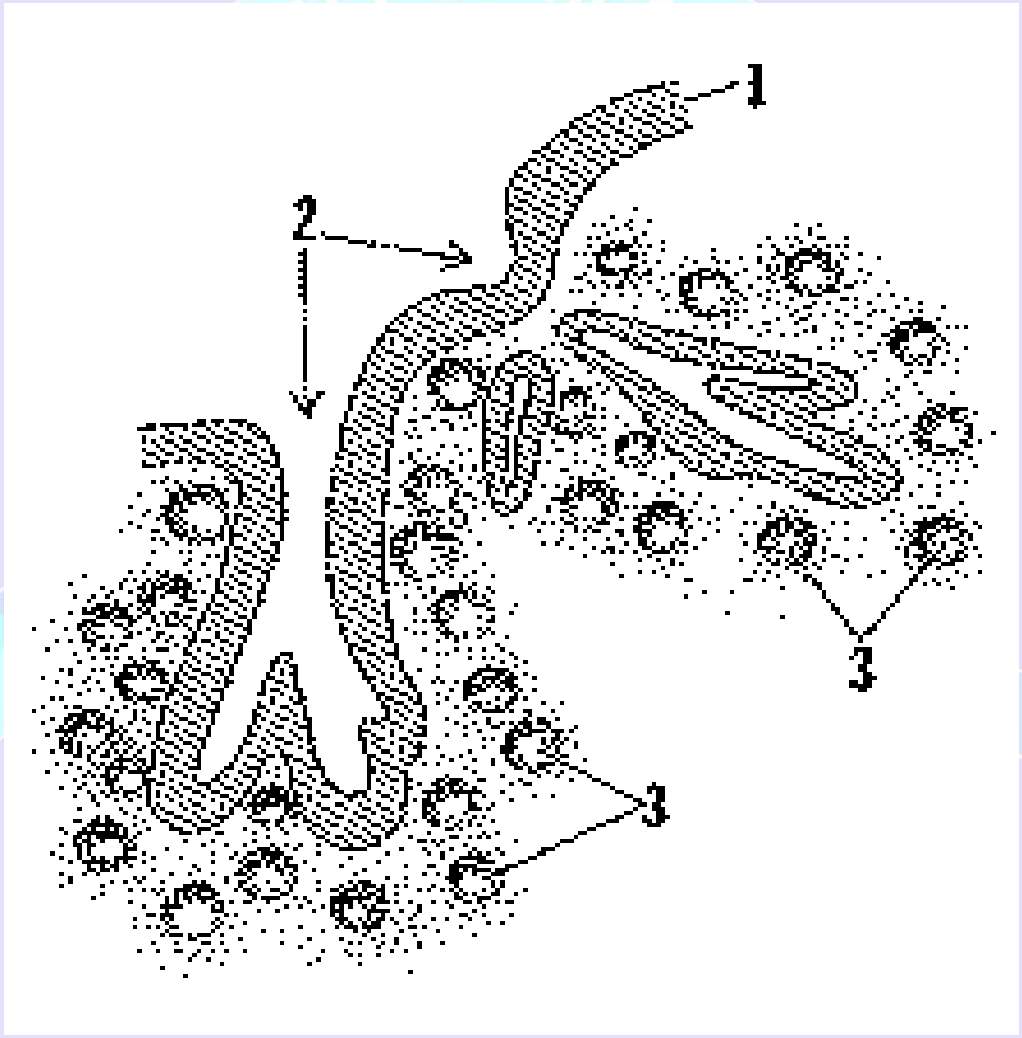
- At various places on the surface of tonsil deep indentations or pockets occur which are known as crypts.

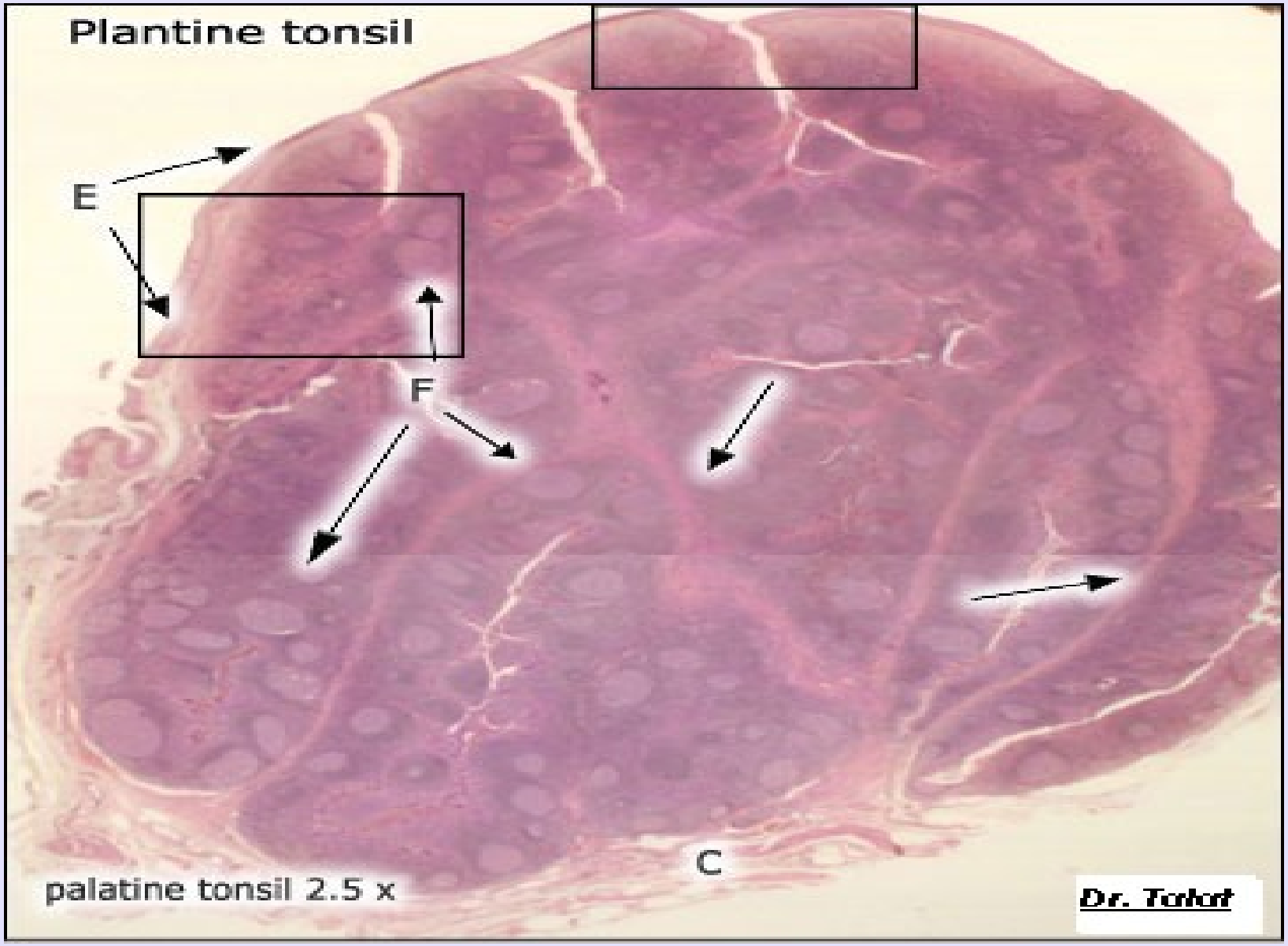
- The crypts are 10~20 in number.



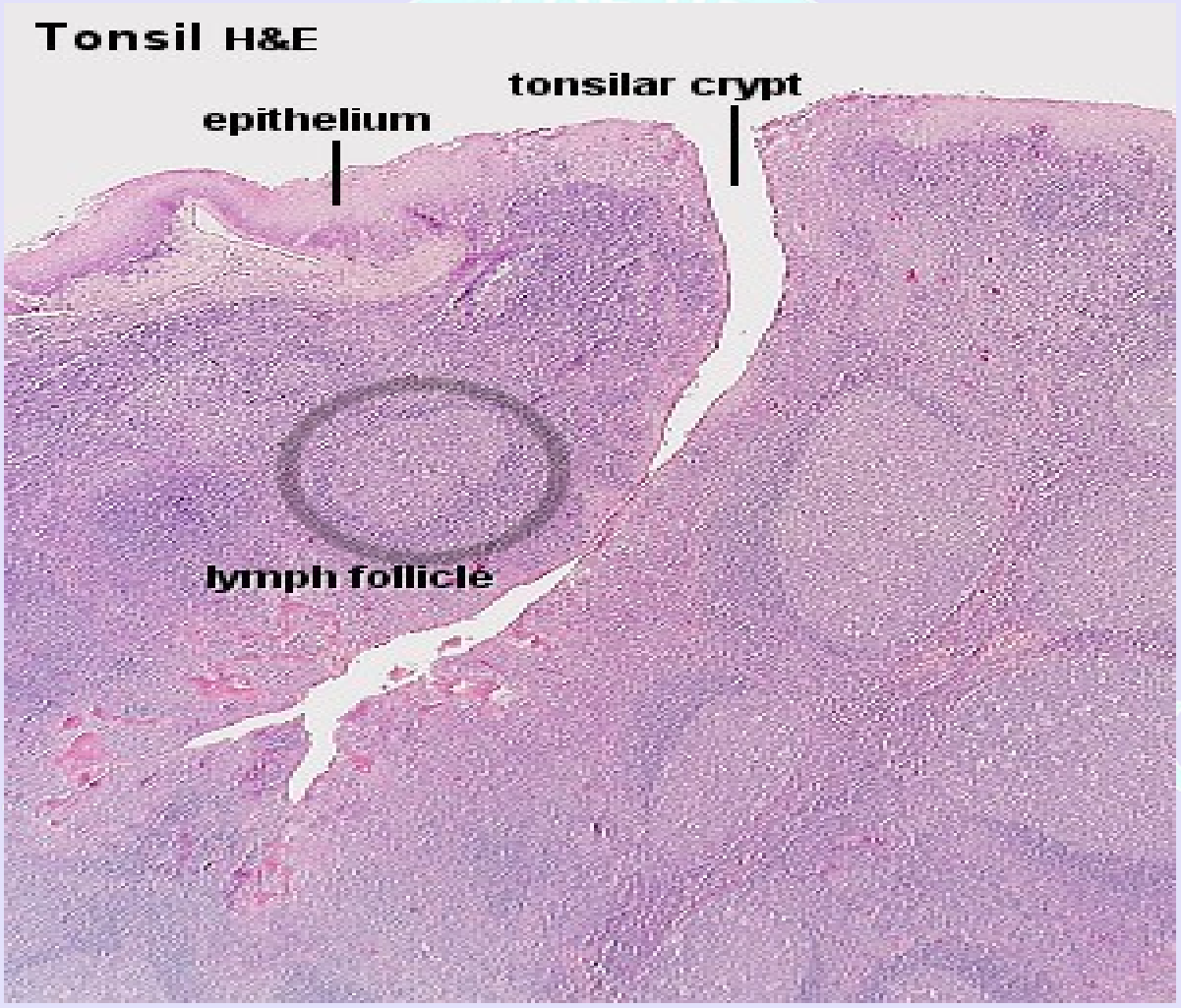
- Several secondary crypts arise from the primary one.
- Surrounding each crypt are lymphatic nodules as well as diffuse lymphatic tissue.
- There is capsule of connective tissue over the attached or basal surface of lymphatic tissue.
- Epithelium of the crypt is infiltrated with lymphocytes.
- In tonsillitis other blood cells like neutrophil are also found in tonsillar tissue

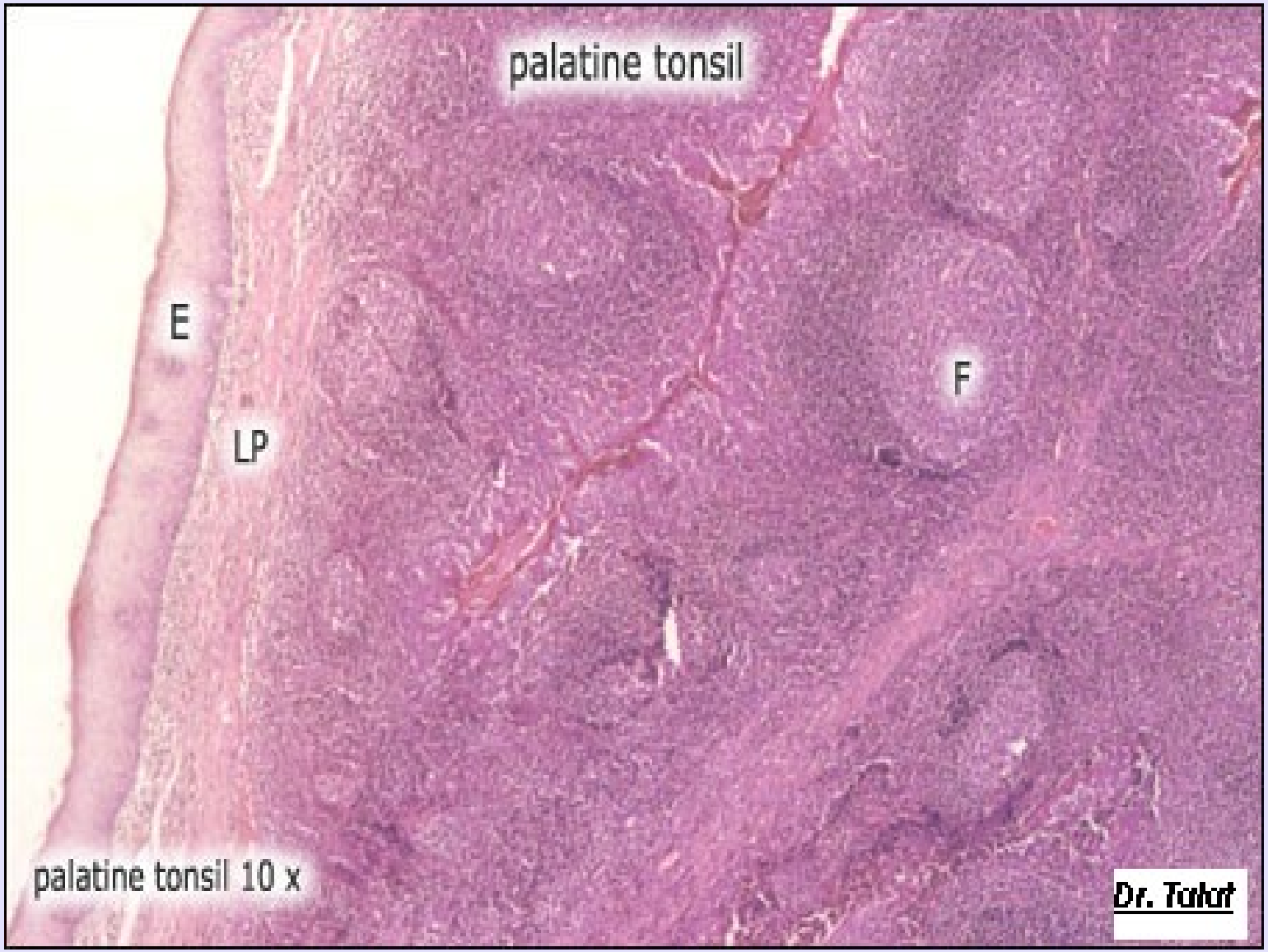
Palatine Tonsil (diagrammatic)




Plantine tonsil
E
F
C
palatine tonsil 2.5 x
Dr. Tahat

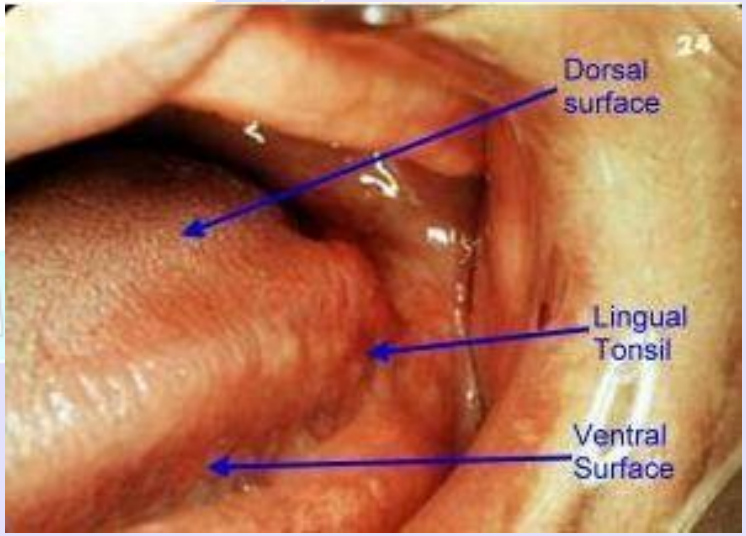
Palatine Tonsil



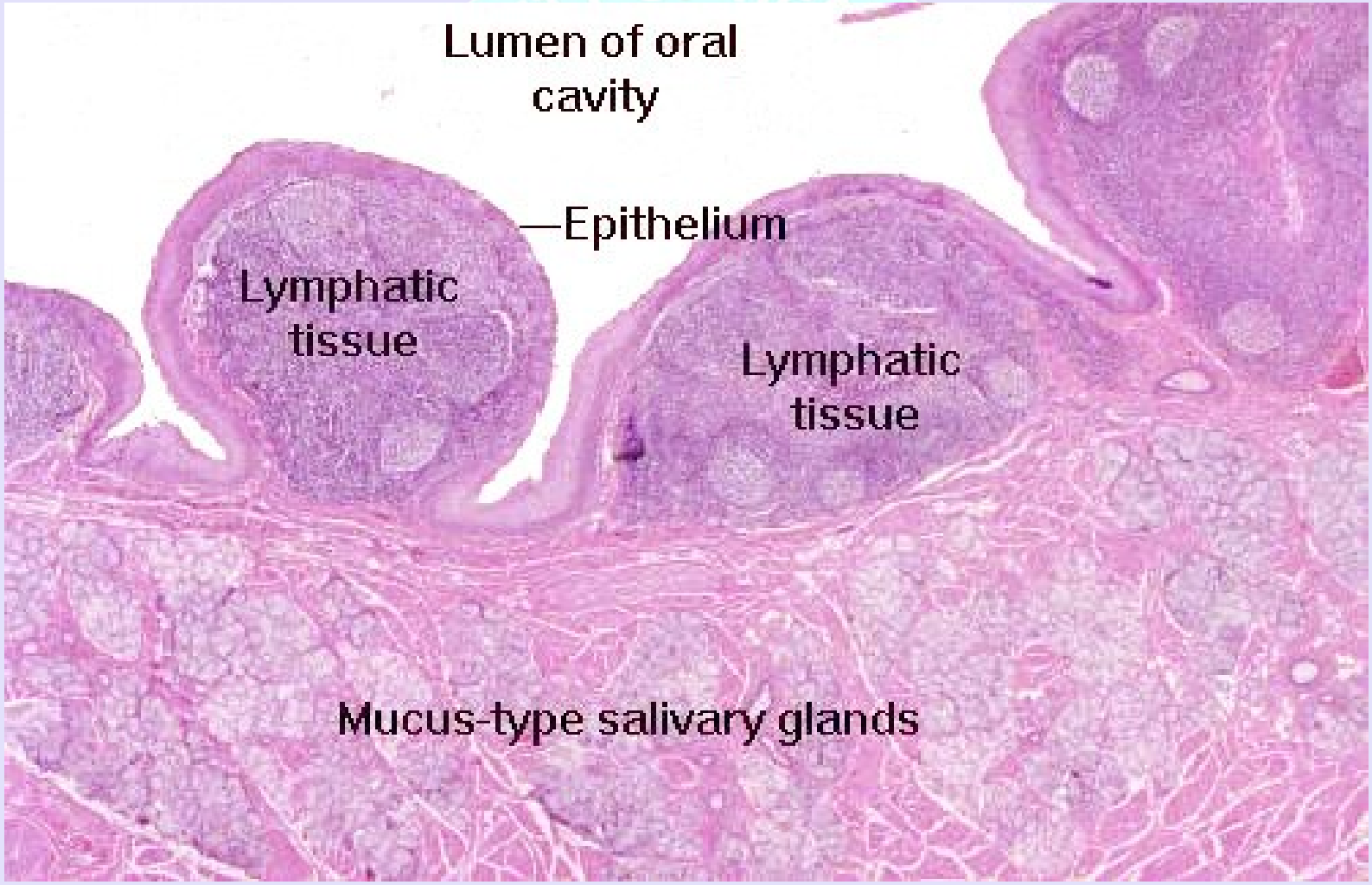

palatine tonsil
E
LP
F
palatine tonsil 10 x
Dr. Takat

Lingual Tonsil

- These are spherical aggregation of lymphatic tissue present on the dorsal surface of tongue.
- Wide mouth crypts are present which are lined by stratified squamous epithelium.
- Ducts of some lingual glands frequently open in these crypts.



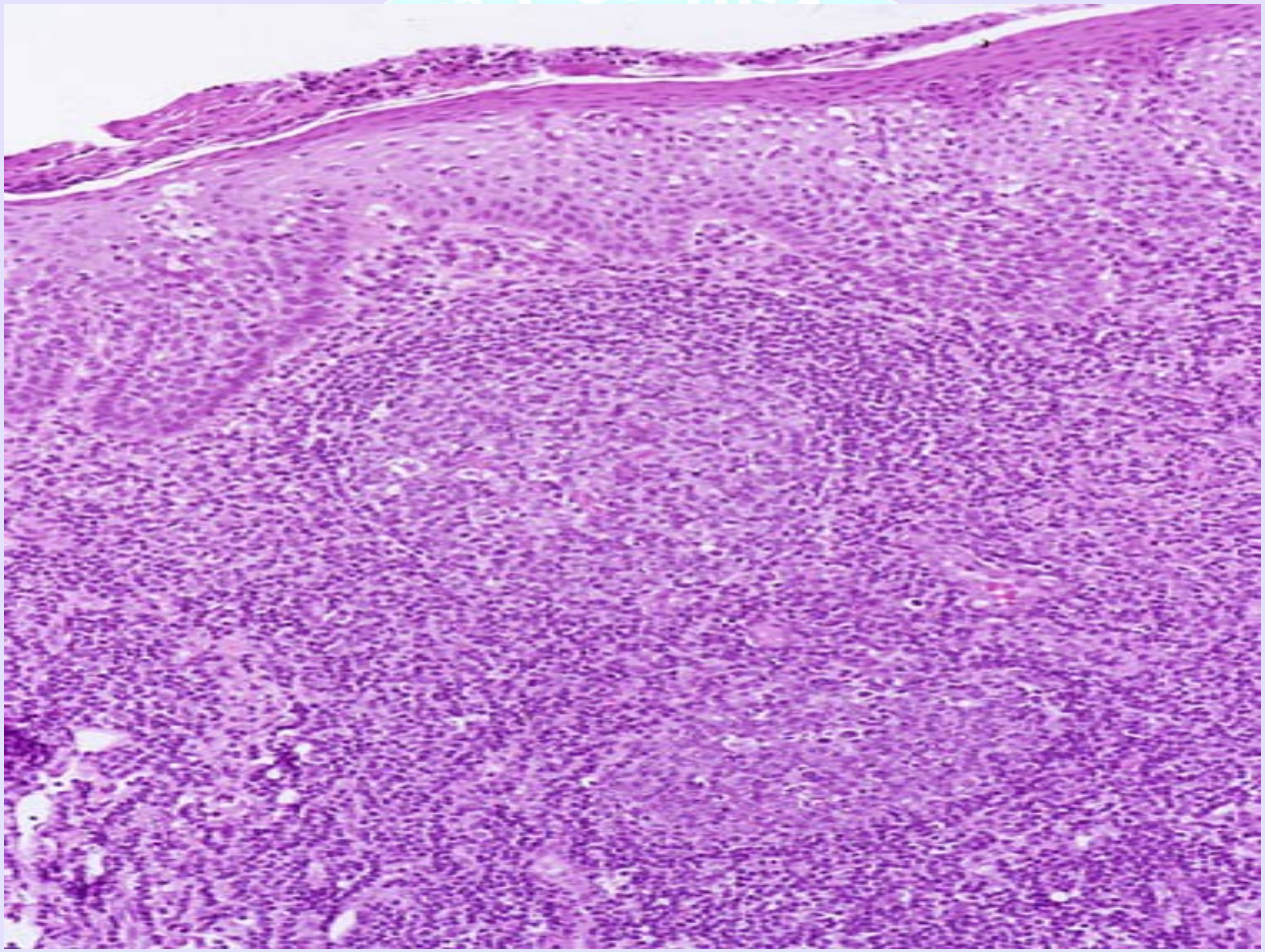
Lingual Tonsil



Pharyngeal Tonsil

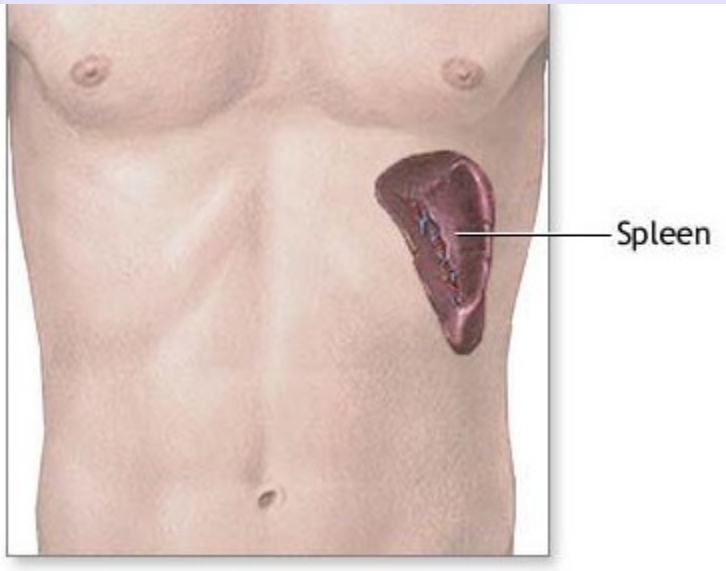
- These are lymphatic aggregation on the posterior wall of nasopharynx.
- Lymphatic tissue is similar to that of palatine tonsil.
- Surface is covered by pseudo-stratified ciliated columnar epithelium.
- In children they are frequently enlarged and obstruct the nasal passages called Adenoids.
- it does not have crypts.

Pharyngeal Tonsil



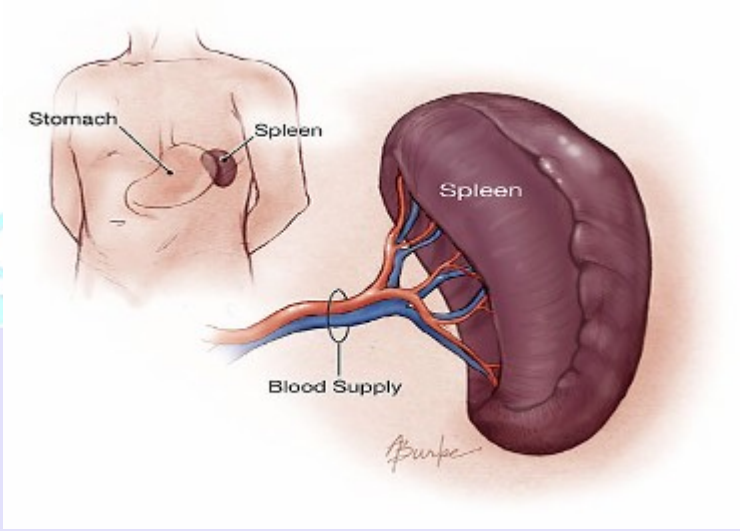
Spleen

- The spleen is the largest lymphoid organ in the body.



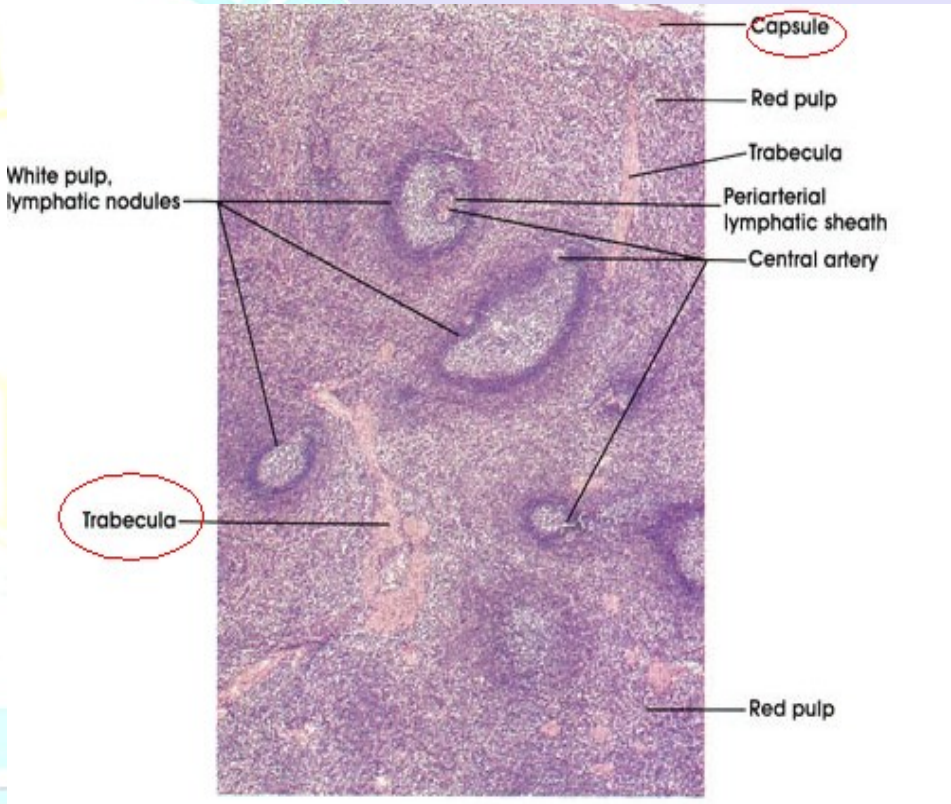
- Only organ specialized for filtering blood.
- It does not have afferent lymphatics.

- Its sinuses are filled with blood instead of lymph.



Spleen

- It is covered by a connective tissue capsule, from which trabeculae extend into substance of organ.

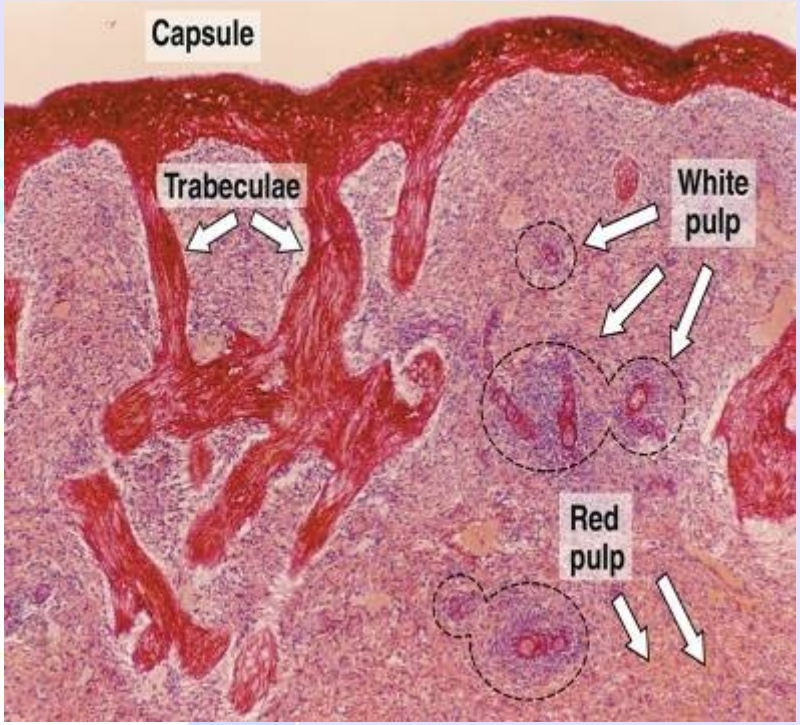


Spleen

- PARENCHYMA or SPLENIC PULP

- Freshly cut unfixed examined by naked eye:
- Red pulp
- White pulp

  Supported by meshwork of R.T.



Section of spleen

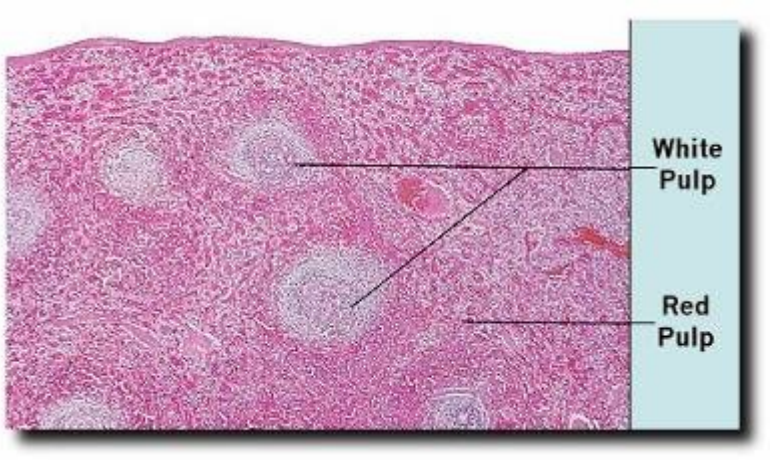
The capsule is seen sending trabeculae to the interior of the organ. The red pulp occupies most of the microscopic field. Note the white pulp with its arterioles.

Histology of spleen

- Histologically, red pulp is "red" due to the presence of large numbers of erythrocytes in blood vessels called sinuses and white pulp is "white" due to lack of these sinuses and consequently fewer erythrocytes.

- The red pulp surrounds the white pulp while the latter looks like lymphatic nodules.

- Closer inspection of the white pulp indicates that there is a "central arteriole", sometimes called a central artery, close to the center of each area of white pulp.



Red pulp

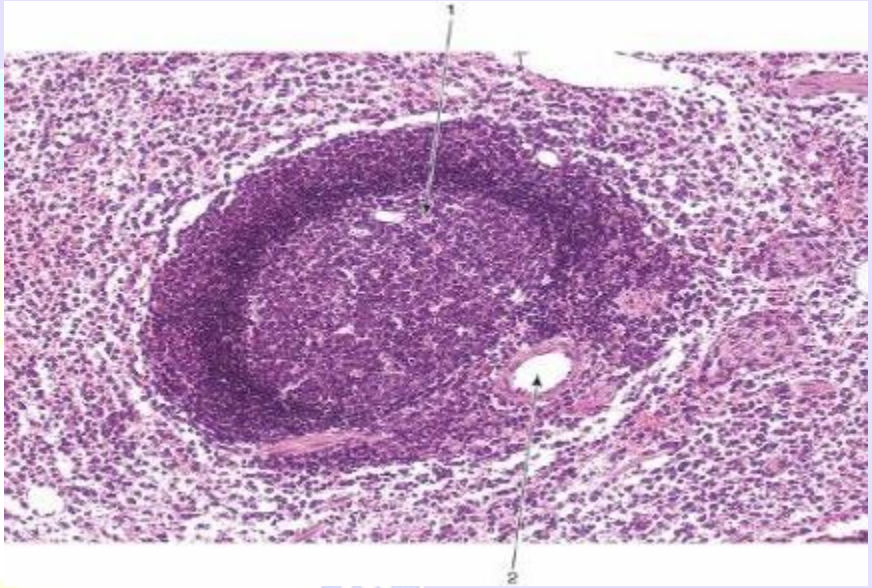
- Under microscope, contain a large number of venous sinuses (sinusoids).
- Splenic or Billroth's cords, contains:
- Splenic sinusoids have irregular lumen & are highly distensible. Their walls contain specialized elongated cells, called "Littoral cells."
- Venous sinus pour their blood into the pulp veins, which unite to form trabecular veins.
- Trabecular veins do not have muscular walls.



White pulp

- It consists of typical lymphatic tissue i.e. densely packed small lymphocytes, that surrounds & follows branches of splenic artery, forming PALS.

- At places, lymphatic sheath thickened to form fusiform or ovoid masses called “splenic nodules or Malpighian bodies”. Which frequently show germinal centers.



Lymphoid nodule of the spleen surrounded by red pulp

A germinative center (1) and the (eccentric) central artery (2), which is characteristic of the spleen, are clearly visible. Two small sections of sheathed arteries are seen to the right of the nodule. H&E

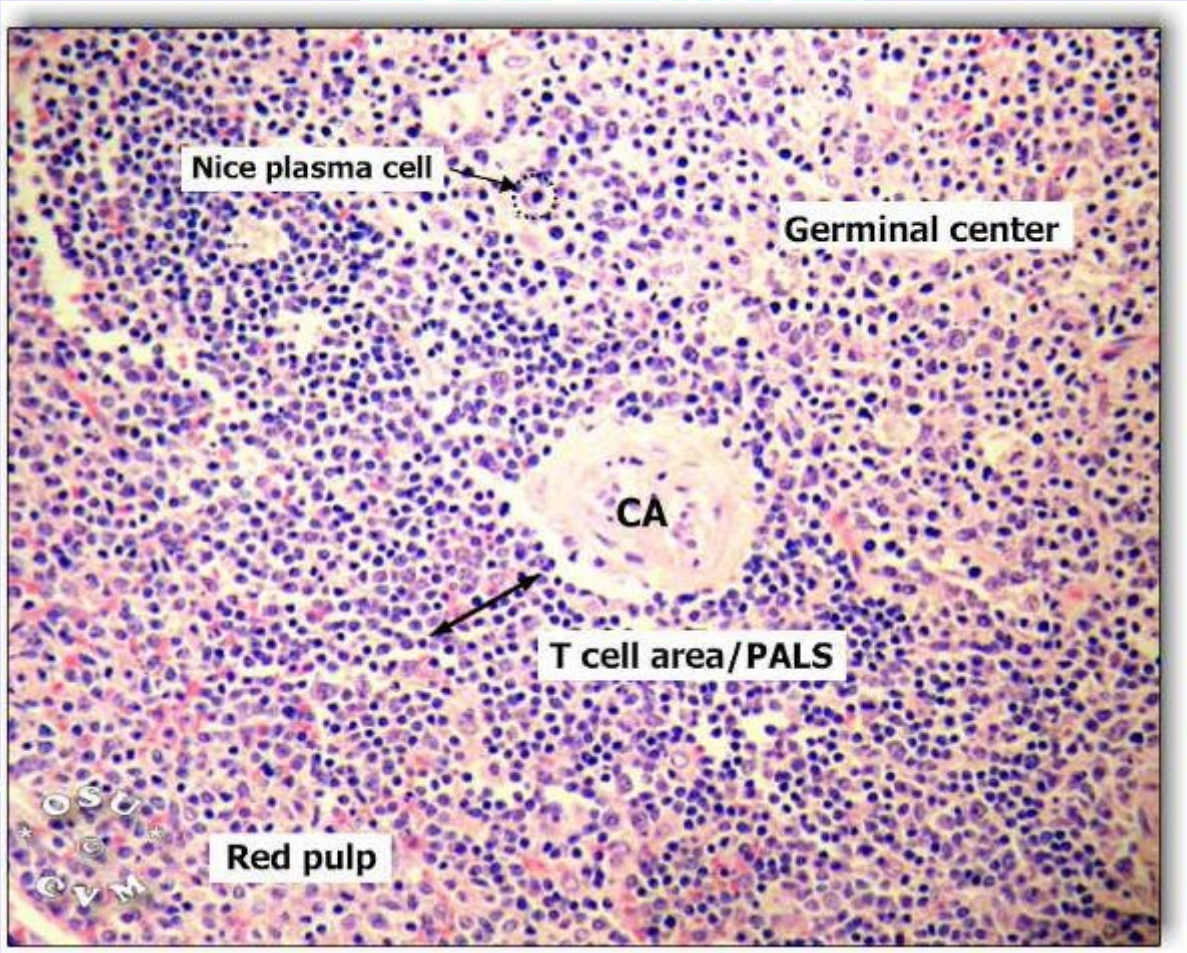
White pulp

- Periarterial lymphatic sheath contains T-lymphocytes, while the splenic nodules lodge B-lymphocytes.

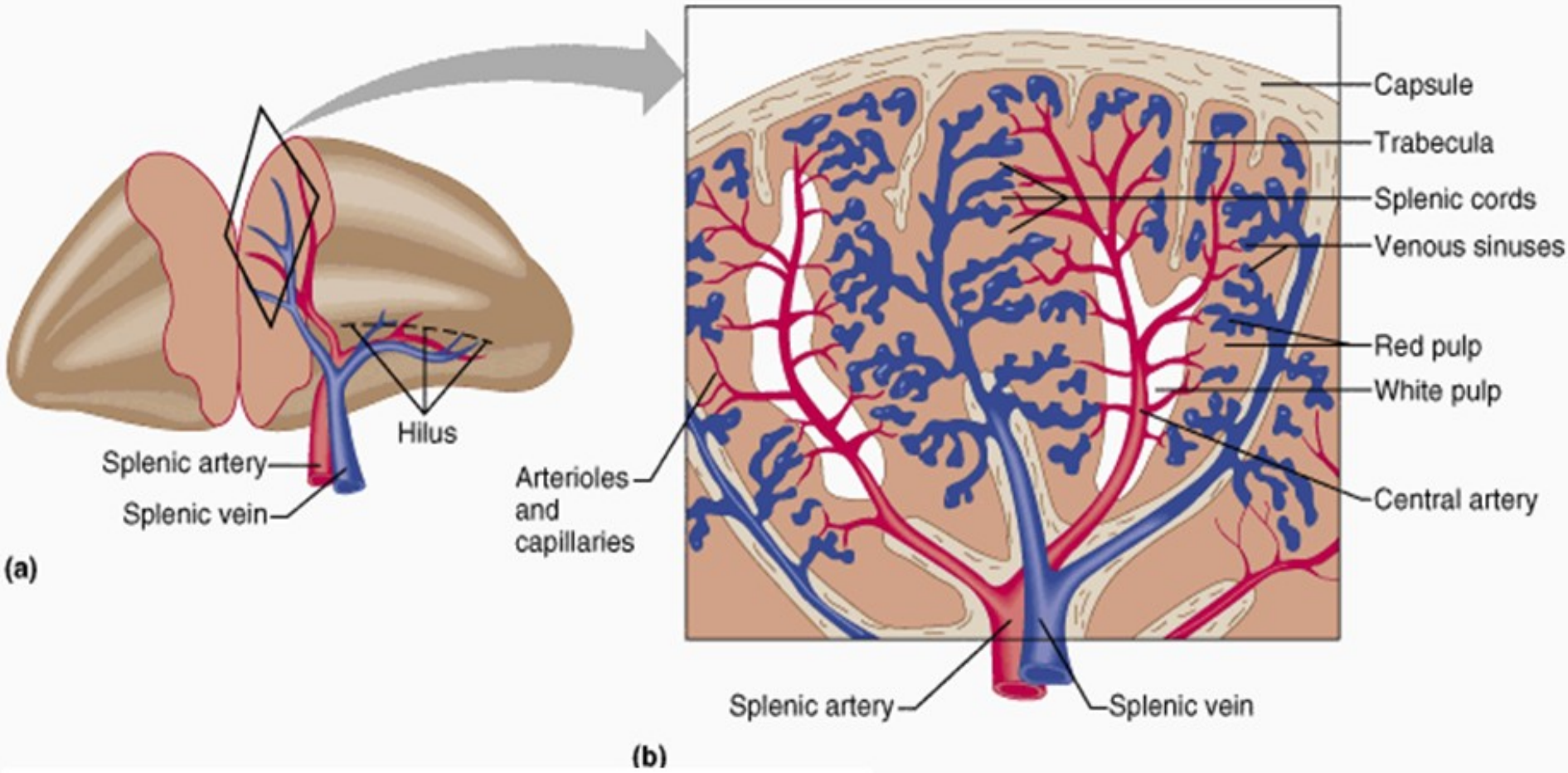
MARGINAL ZONE

- Loose LT– many small venous sinuses.
- Few lymphocytes but many active macrophages.
- Here circulating antigens are trapped. Play imp. role in filtration of blood, and immunological activity of the spleen.

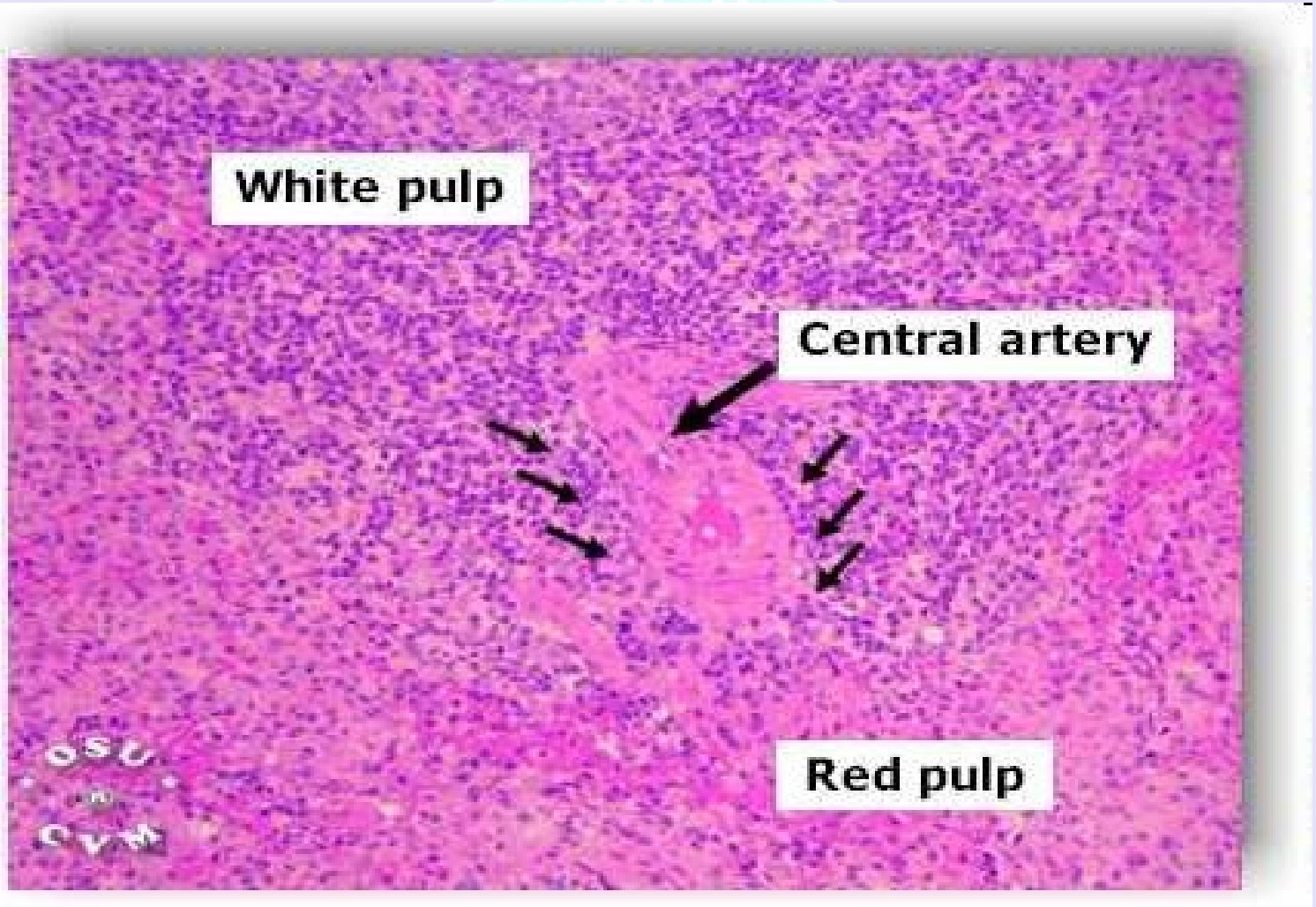
Red and white pulp



Blood Circulation



Spleen



Functions of Spleen

- Filtration of blood: Spleen reacts promptly to remove the antigen carried by blood.
- Production of lymphocytes: occurs in white pulp, migrate to red pulp and reach the lumen of the sinusoids, where they incorporated into the blood.
- Destruction of old & worn out erythrocytes. Carried out by macrophages of the red pulp.
- Storage of blood. due to its spongy structure.
- In human, storage capacity is very small.

References

- Wheater’s Functional Histology. (A Text and color Atlas).

- Text/Atlas of Histology by Leeson and Paparo.